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CERTIFICATE

This is to certify that the case study entitled “**Sustainable Energy Solutions: Addressing the Growing Demand through Solar Power**” is a bonafide work carried out by {**Mubashira K M, Mumtaz Begum, N Merlyn Sahana, P Shireen Naaz**} in partial fulfillment for the award of degree of **Bachelor Degree in Computer Science & Engineering** in the VISVESVARAYA TECHNOLOGICAL UNIVERSITY, Belagavi during the academic year 2025-2026. It is certified that all corrections and suggestions indicated for internal assessment have been incorporated in the report deposited in the library. The case study has been approved as it satisfies the academic requirements prescribed for a Bachelor of Engineering Degree.

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Abstract

Solar energy is a rapidly growing renewable energy source that offers a sustainable solution to the increasing global demand for electricity. This report presents an overview of solar energy, including its importance, working principle, implementation methodology, and performance analysis. Solar power systems utilize the photovoltaic effect to convert sunlight into electrical energy, providing a clean and environmentally friendly alternative to fossil fuels.

The study highlights the need for adopting solar energy due to the depletion of conventional energy resources, rising carbon emissions, and lack of reliable electricity in remote areas. It also explains the process of designing, installing, and maintaining solar energy systems to ensure optimal efficiency. Data analysis shows that solar systems are economically viable, with reasonable installation costs, low maintenance, and a payback period of 4–6 years.

Despite certain limitations such as high initial investment and weather dependency, advancements in solar technology and government support are driving widespread adoption. The report concludes that solar energy plays a crucial role in achieving sustainable development and reducing environmental impact, making it a key component of future energy systems.

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